

Asdfgh JKL;Ex. No 4: Analyze email headers and detect email spoofing using MHA (Mail Header Analyzer)

Description:

Here's a step-by-step guide on how to execute and analyze an email header:

1. Access the Email Header

- **Gmail:** Open the email, click the three dots (More) in the upper right corner, and select "Show original."
- **Outlook:** Open the email, click on "File," then "Properties," and find the "Internet headers" box.
- **Yahoo:** Open the email, click on the three dots (More), and select "View raw message."

2. Copy the Email Header

- Once you access the header, copy all the text that appears. This text contains various metadata fields about the email's journey from the sender to your inbox.

3. Identify Key Header Fields

- **From:** The sender's email address.
- **To:** The recipient's email address.
- **Date:** The date and time the email was sent.
- **Subject:** The email's subject line.
- **Return-Path:** The return address if the email bounces.
- **Received:** A list of servers the email passed through on its way to the recipient. This is usually the most critical part of header analysis.
- **Message-ID:** A unique identifier for the email.
- **SPF (Sender Policy Framework) / DKIM (DomainKeys Identified Mail):** Indicators of email authentication.

4. Analyze the 'Received' Fields

- The Received fields show the path the email took from the sender to the recipient. These fields are listed in reverse order (from the last server to the first).
- Each Received line will show:
 - **The sending server's hostname/IP address.**
 - **The receiving server's hostname/IP address.**
 - **The date and time the email passed through.**

5. Check for IP Addresses and Hostnames

- Use tools like **WHOIS** or online IP lookup services to identify the geographical location and ownership of the IP addresses found in the Received lines.

- Check if any IP addresses are suspicious or if the hostname does not match the expected sending server.

6. Examine the SPF, DKIM, and DMARC Results

- **SPF:** Checks if the sender's IP is authorized to send emails for the domain.
 - **Pass:** Indicates the IP is authorized.
 - **Fail:** The IP is not authorized and could be a sign of spoofing.
- **DKIM:** Verifies that the email content has not been altered in transit.
 - **Pass:** The email has not been tampered with.
 - **Fail:** The email content may have been altered.
- **DMARC:** A policy that uses SPF and DKIM to authenticate the email.
 - **Pass:** Indicates alignment with SPF/DKIM policies.
 - **Fail:** Indicates possible spoofing or tampering.

7. Analyze the 'Message-ID'

- The Message-ID is unique to each email. It usually includes the domain of the sending server.
- Check if the domain in the Message-ID matches the sender's domain. If not, it could be a sign of spoofing.

8. Look for Anomalies

- **Mismatch in Email Domains:** Check if the From domain matches the domains in Return-Path, Message-ID, and other authentication results.
- **Strange IP Addresses:** Be wary of IP addresses that do not match the known sending server or are from unexpected locations.
- **Suspicious Time Stamps:** Check for significant time discrepancies in the Received lines, which could indicate delays or rerouting.

9. Use Online Tools

- Use online email header analysis tools like **MXToolbox** or **Google's G Suite Toolbox** to automatically parse and analyze the email header.

10. Document and Report Findings

- Document all your findings, especially if you detect phishing or spoofing.
- Report suspicious emails to your IT department or email provider for further investigation.

Example of a Header Analysis

Here's a quick example with a sample header:

vbnet

Copy code

Received: from mail.example.com (mail.example.com [192.0.2.1])

by mail.receiver.com with ESMTP id u29si8604336pjs.40.2023.08.10.07.00.16;

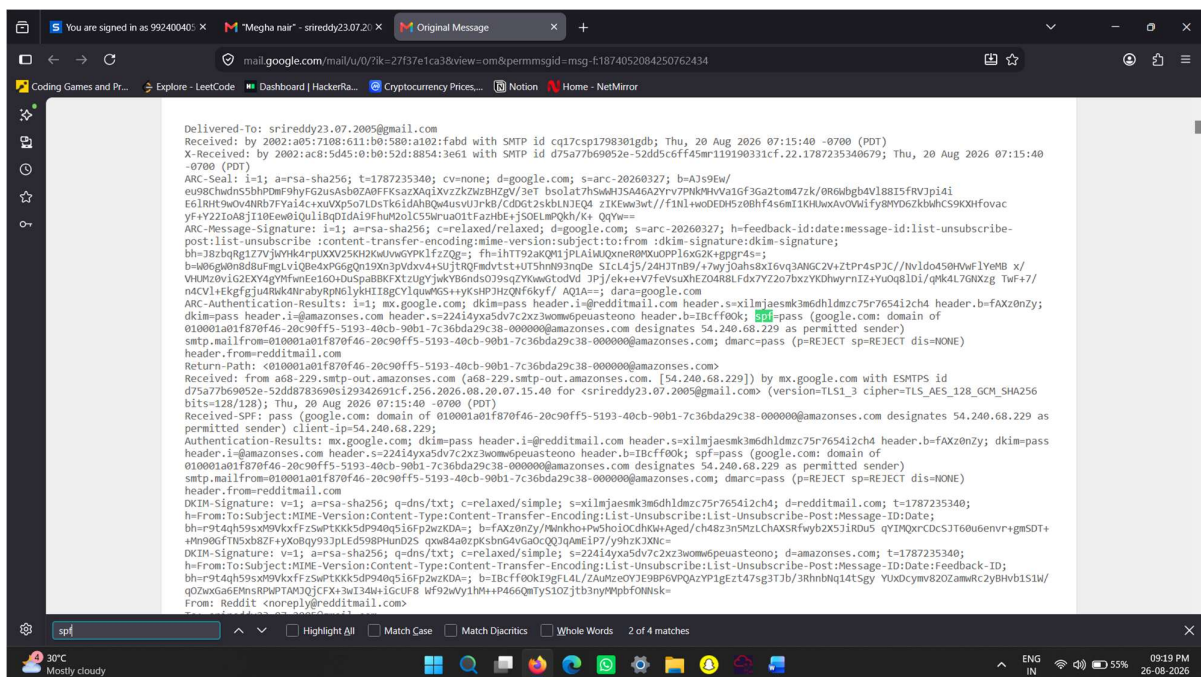
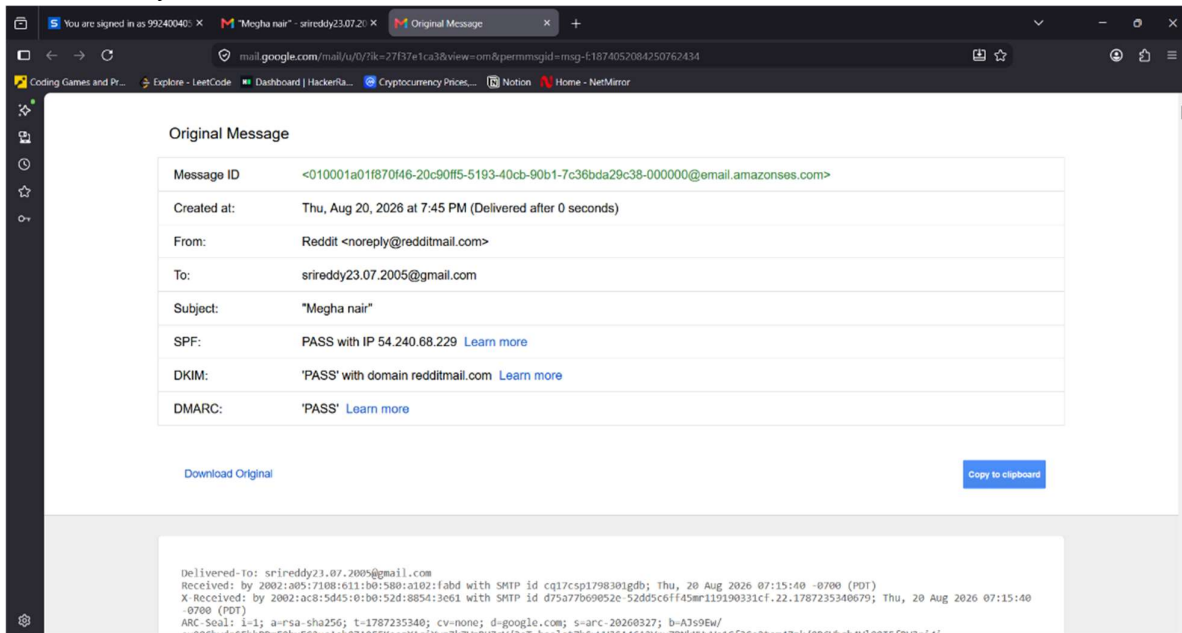
Thu, 10 Aug 2023 07:00:16 -0700 (PDT)

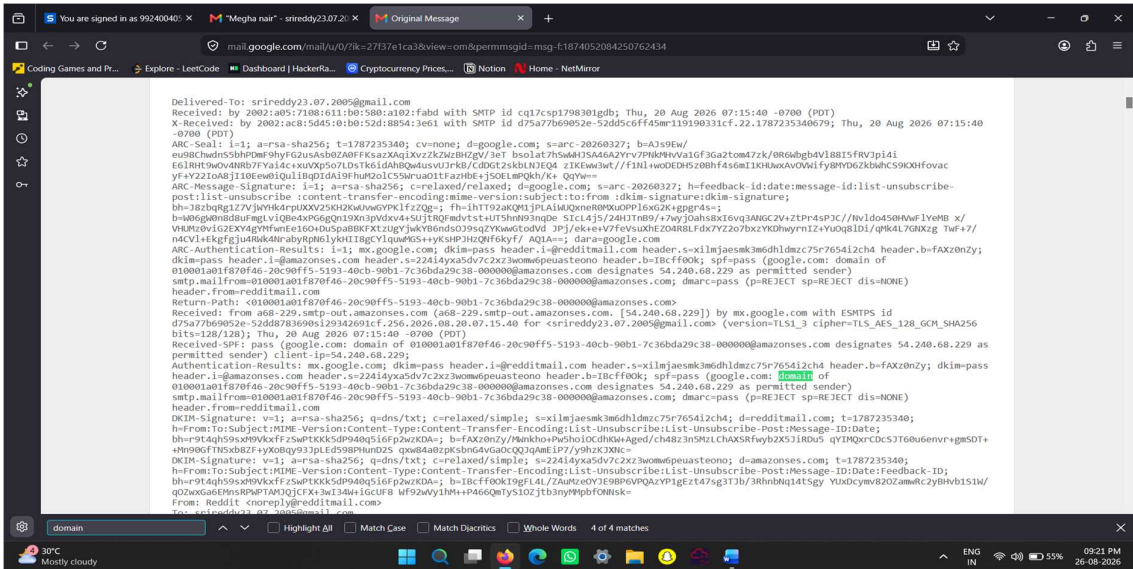
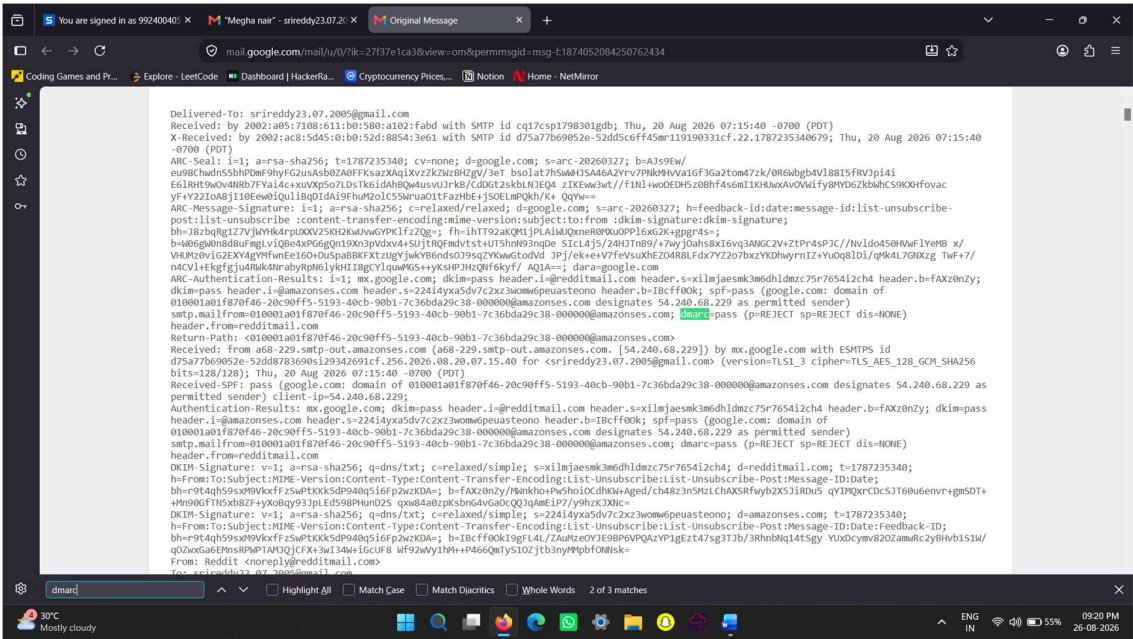
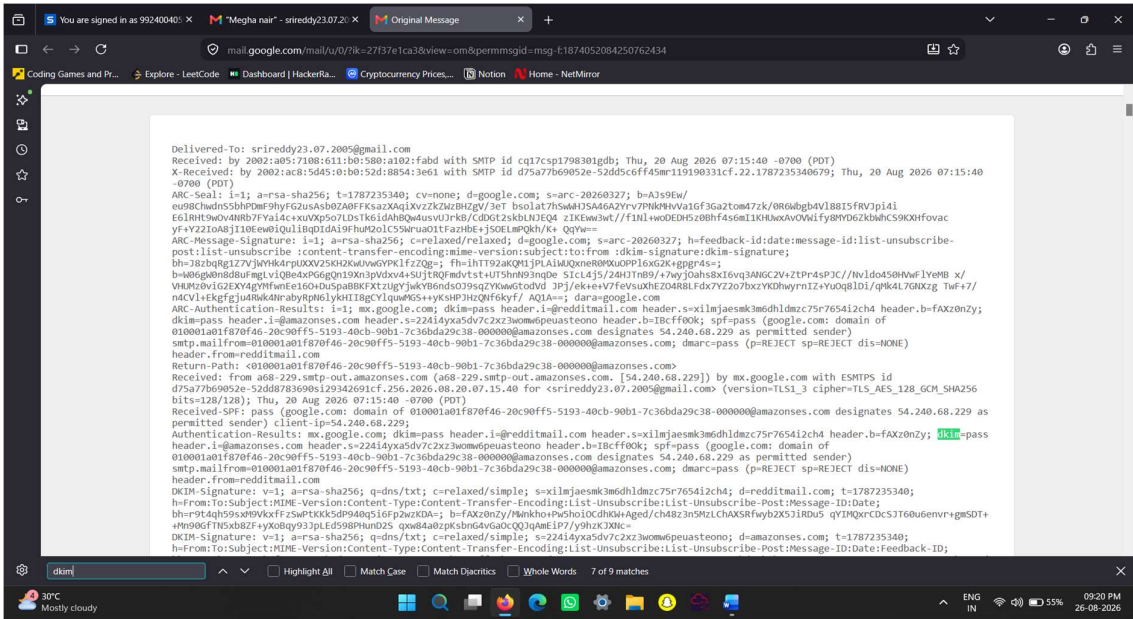
Received: by mail.example.com with SMTP id almr1243772ywh.51;

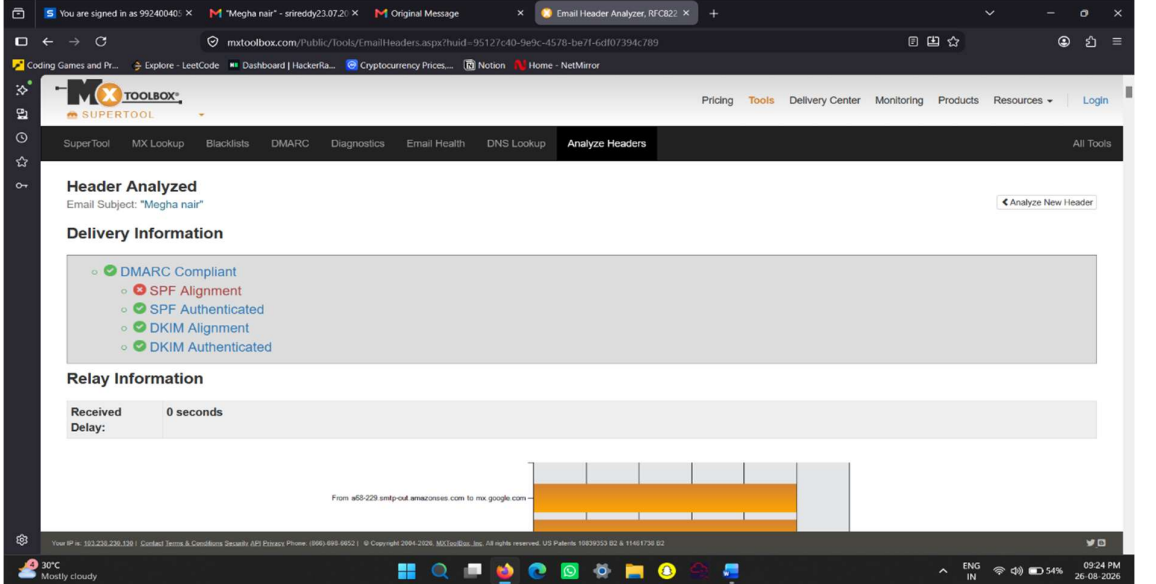
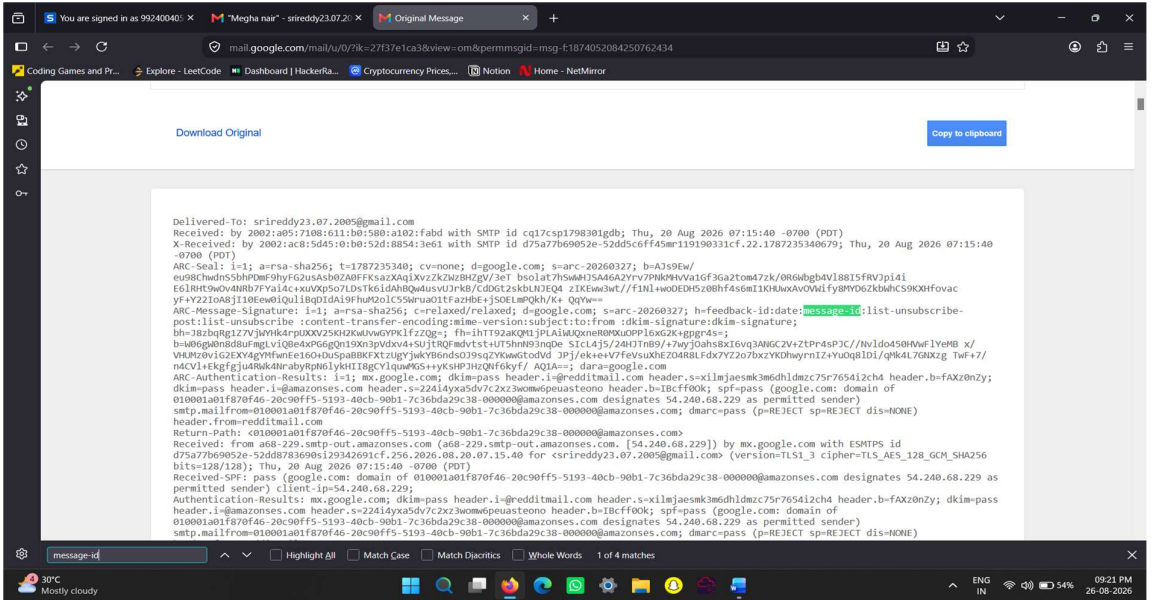
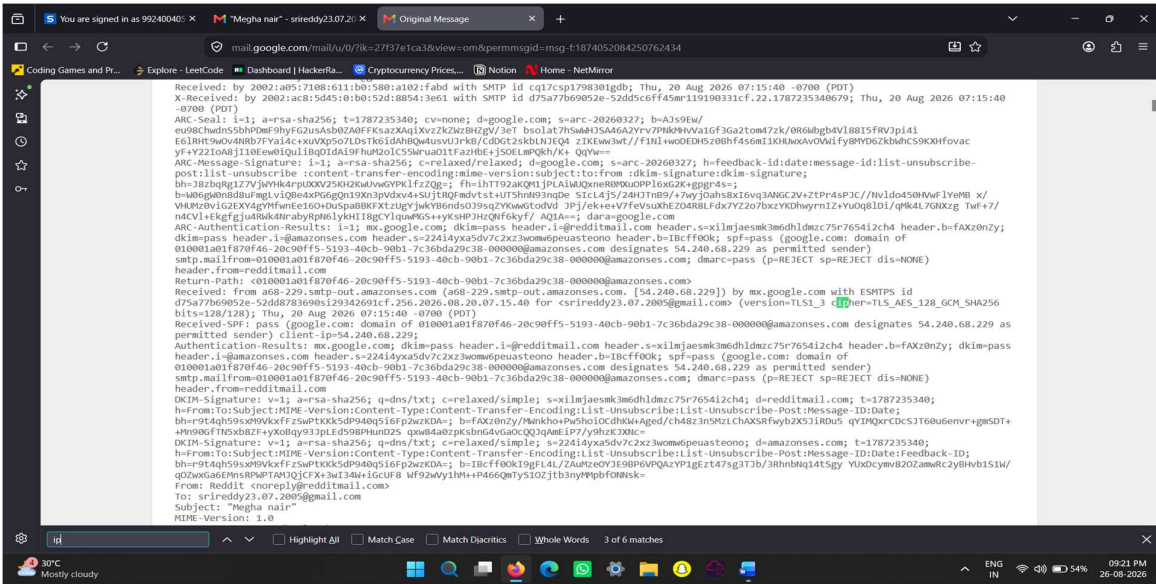
Thu, 10 Aug 2023 07:00:15 -0700 (PDT)

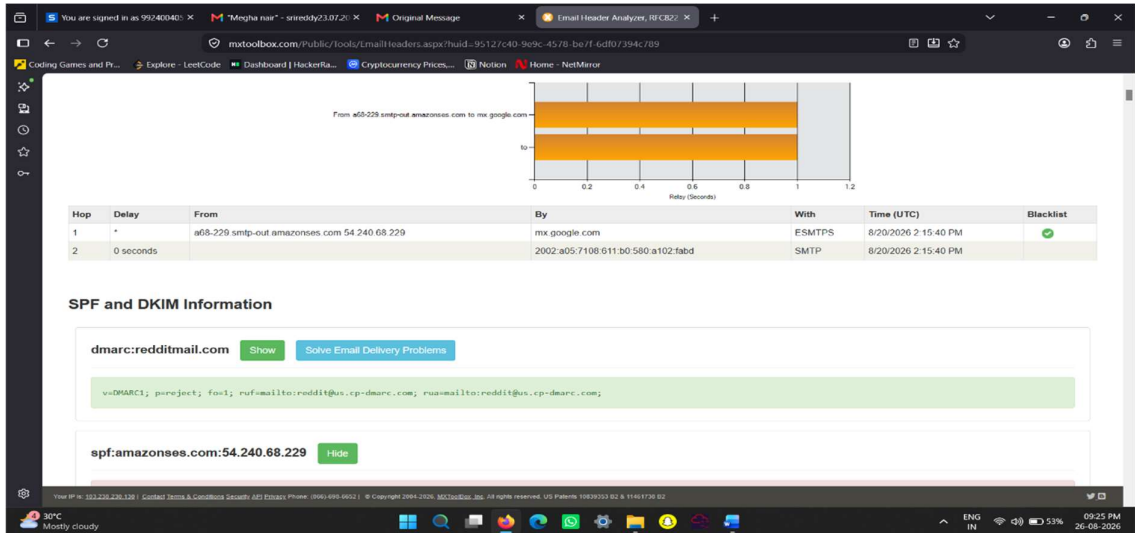
Message-ID: <CA+7eu=4pSeXgQ@mail.example.com>

- **Check IP Addresses:** Lookup 192.0.2.1 and ensure it is associated with mail.example.com.
- **Verify SPF/DKIM:** Check if mail.example.com is authorized to send emails for this domain.
- **Compare Time Stamps:** Ensure the times make sense chronologically without unusual delays.









v=spf1 ip4:199.255.192.0/22 ip4:199.127.232.0/22 ip4:54.240.0.0/18 ip4:69.169.224.0/20 ip4:23.249.208.0/20 ip4:23.251.224.0/19 ip4:76.223.176.0/20 ip4:54.240.64.0/18 ip4:76.223.128.0/19

Prefix	Type	Value	PrefixDesc	Description
v		spf1		The SPF record version
+	ip4	199.255.192.0/22	Pass	Match if IP is in the given range.
+	ip4	199.127.232.0/22	Pass	Match if IP is in the given range.
+	ip4	54.240.0.0/18	Pass	Match if IP is in the given range.
+	ip4	69.169.224.0/20	Pass	Match if IP is in the given range.
+	ip4	23.249.208.0/20	Pass	Match if IP is in the given range.
+	ip4	23.251.224.0/19	Pass	Match if IP is in the given range.
+	ip4	76.223.176.0/20	Pass	Match if IP is in the given range.
+	ip4	54.240.64.0/18	Pass	Match if IP is in the given range.
+	ip4	76.223.128.0/19	Pass	Match if IP is in the given range.
+	ip4	210.221.160.0/19	Pass	Match if IP is in the given range.
+	ip4	208.55.144.0/20	Pass	Match if IP is in the given range.
+	ip4	24.110.64.0/18	Pass	Match if IP is in the given range.
+	ip4	96.77.0.0/16	Pass	Match if IP is in the given range.
-	all		Fail	Always matches. It goes at the end of your record.
From Domain		redditmail.com		The domain used in the From header field.
Return Path Domain		amazonses.com		The domain used in the Return-Path header field.

Test	Result	More Info
SPF Alignment	Domain not found in SPF	More Info
SPF Record Published	SPF Record found	
SPF Record Deprecated	No deprecated records found	
SPF Multiple Records	Less than two records found	
SPF Contains characters after ALL	No items after 'ALL'	
SPF Syntax Check	The record is valid	
SPF Included Lookups	Number of included lookups is OK	
SPF Recursive Loop	Nor Recursive Loops on Includes	
SPF Duplicate Include	No Duplicate includes Found	
SPF Type PTR Check	No type PTR found	
SPF Void Lookups	Number of void lookups is OK	
SPF MX Resource Records	Number of MX Resource Records is OK	
SPF Record Null Value	No Null DNS Lookups found	
SPF Authentication	SPF Passed for IP	

Your DNS hosting provider is "Amazon Route 53" [Need Bulk Dns Provider Data?](#)

Reported by ns-1722.awsdns-23.co.uk on 8/26/2026 at 3:55:36 PM (UTC 0), just for you. [Transcript](#)

dkim:redditmail.com:xilmjaesmk3m6dhldmzc75r7654i2ch4 [Show](#)

Browser window showing Email Header Analyzer results for a message from srireddy23.07.20@gmail.com. The interface includes a sidebar with navigation options like 'Explore', 'Dashboard', and 'HackerRa...'. The main content area displays the raw email headers, including DKIM signatures and a public record link. Below the headers, a 'Headers Found' section lists the extracted header names and their values, such as 'Delivered-To', 'X-Received', and 'ARC-Seal'.

Browser window showing the 'Headers Found' section of the Email Header Analyzer. This section provides a detailed list of headers and their values, including 'Delivered-To', 'X-Received', 'ARC-Seal', 'ARC-Message-Signature', 'ARC-Authentication-Results', 'Return-Path', 'Received-SPF', 'Authentication-Results', 'DKIM-Signature', 'From', 'To', 'Subject', and 'MIME-Version'. The values are formatted to show the full details of each header, including domain names and timestamps.

Browser window showing the 'Received Header' section of the Email Header Analyzer. This section displays the 'Received' header, which contains information about the message's origin, including the sender's email address, the recipient's email address, and the date and time of receipt. The header is formatted to show the full details of the message's origin.